

Scenario

Cloud computing has become an important technology for storing and managing data securely. Traditional storage methods such as hard disks and local servers have limited storage capacity and are at risk of data loss due to hardware failure. To overcome these challenges, cloud platforms such as Microsoft Azure provide secure and reliable storage solutions.

This project focuses on implementing a cloud file storage system using Microsoft Azure. The infrastructure includes a Resource Group, Storage Account, and Blob Storage Container. These services allow users to upload, store, and manage files securely in the cloud.

The project demonstrates how Microsoft Azure provides a simple, scalable, and cost-effective solution for cloud-based file storage.

Literature Review

The project is based on the practical implementation of Microsoft Azure cloud storage services. Cloud storage is widely used for secure data management and remote accessibility.

The project concept was studied using:

- Microsoft Azure Official Documentation
- Microsoft Learn Platform
- Cloud Computing Study Material
- Azure Portal Practical Implementation

Project Description

The project implements a cloud file storage system using Microsoft Azure.

Services Used

- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage

Workflow

1. Create a Resource Group.
2. Create a Storage Account.
3. Create a Blob Container.

4. Upload files to Blob Storage.
5. Verify uploaded files.

Objectives

- Secure cloud storage.
- Easy file management.
- Remote accessibility.
- Cost-effective storage solution.

Results and Screenshots

The cloud storage system was successfully implemented.

Implemented Resources

- Resource Group created.
- Storage Account configured.
- Blob Container created.
- Files uploaded successfully.

IMPLEMENTATION STEPS

Step 1 — Create Resource Group

Description

A Resource Group named **StorageProjectRG** was created to manage Azure resources.

Output

Resource Group created successfully.

Figure 1:

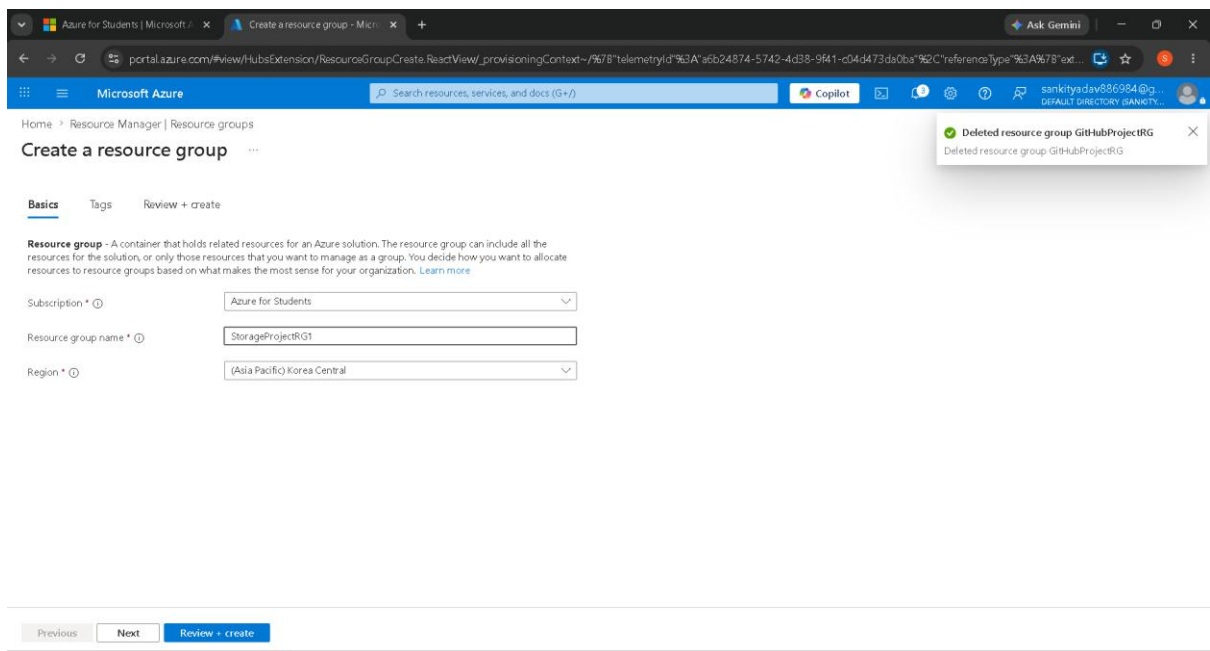


Figure 2:

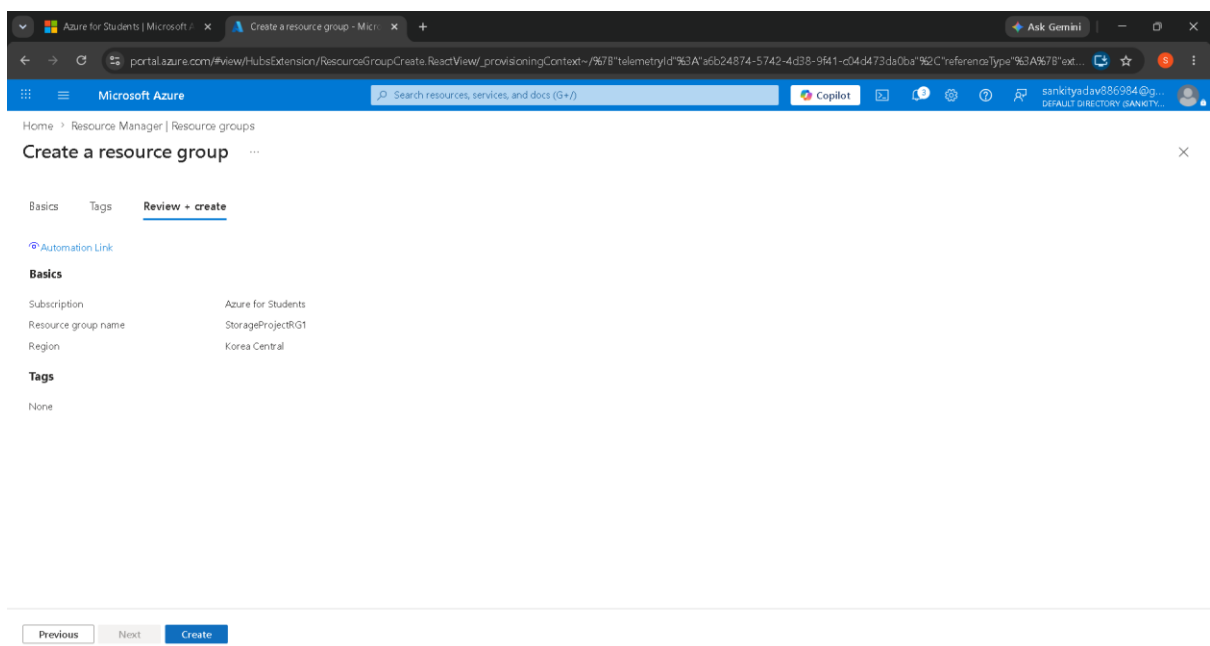
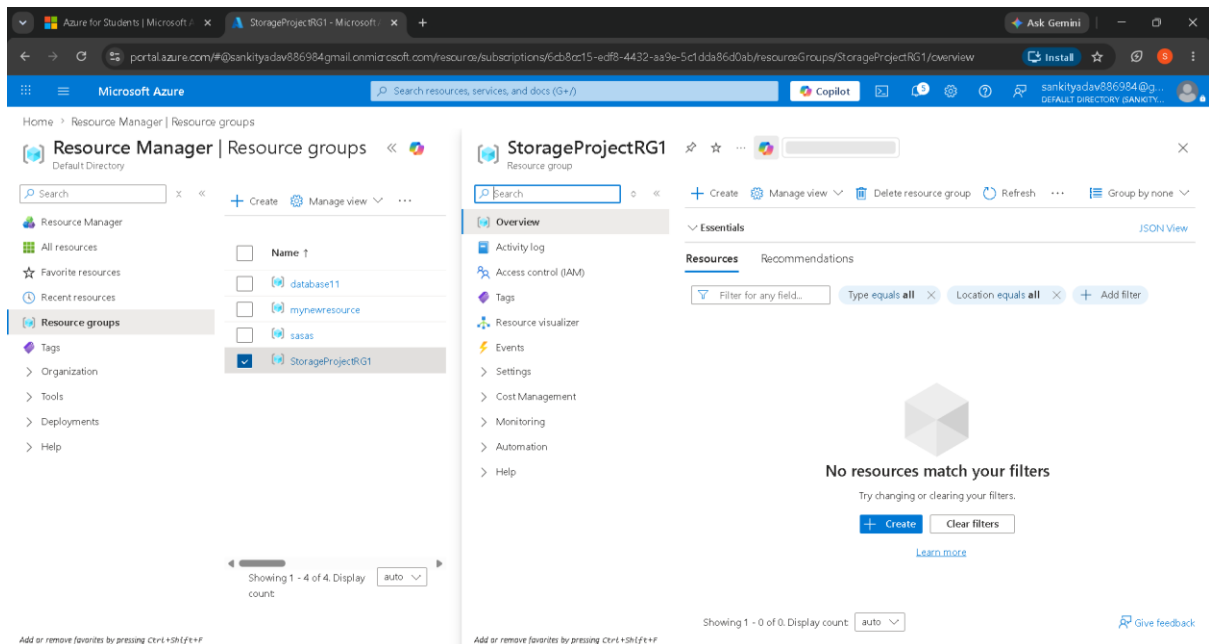


Figure 3:



Step 2 — Create Storage Account

Description

A Storage Account was created to securely store cloud files.

Output

Storage Account created successfully.

Figure 4:

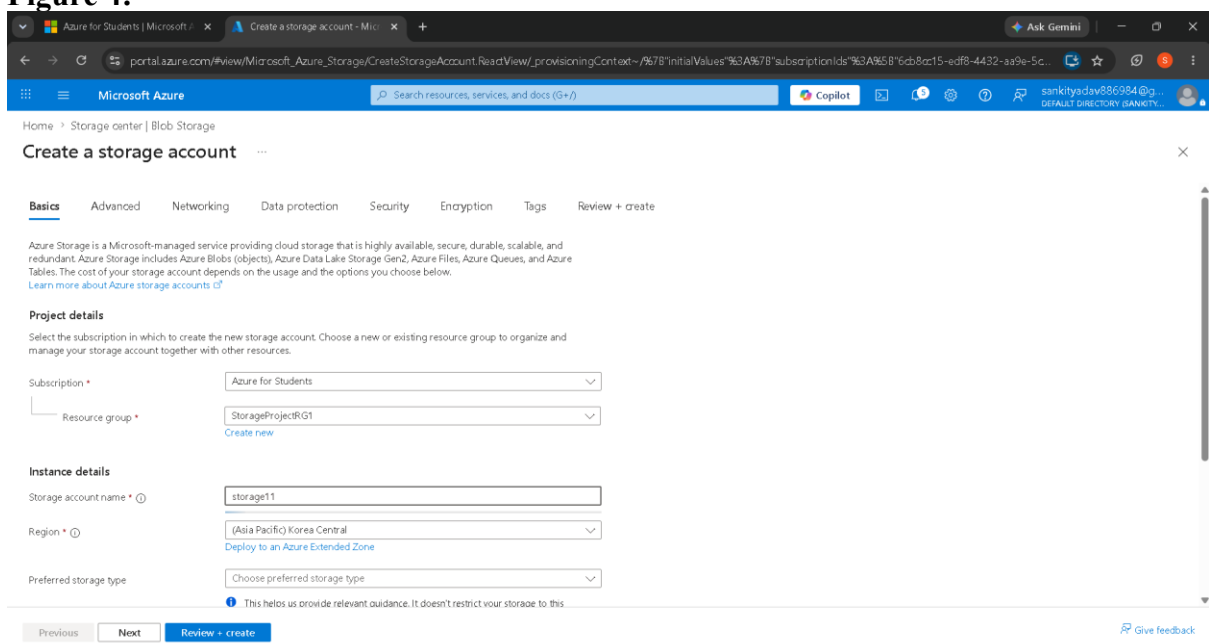


Figure 5:

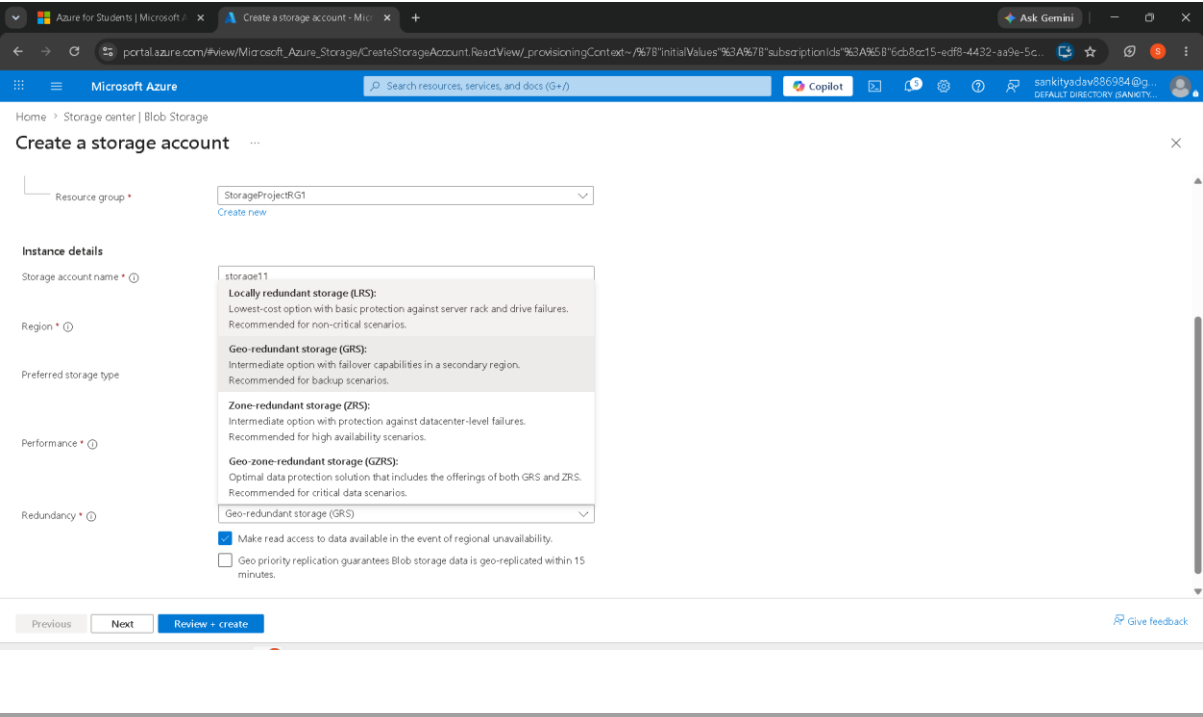


Figure 6:

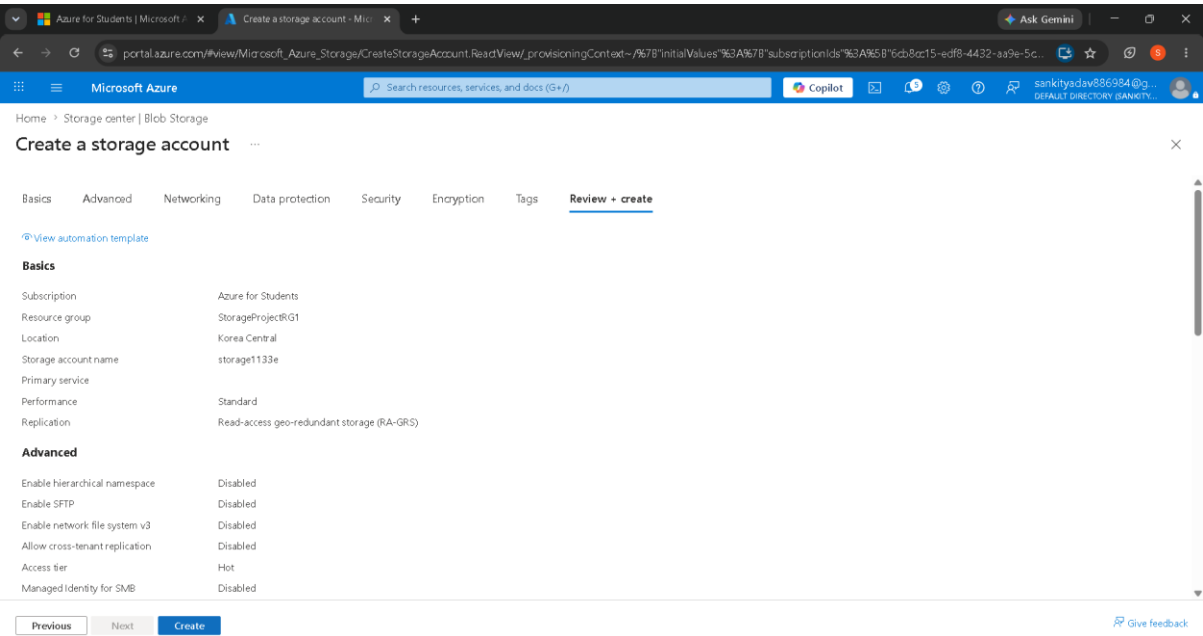
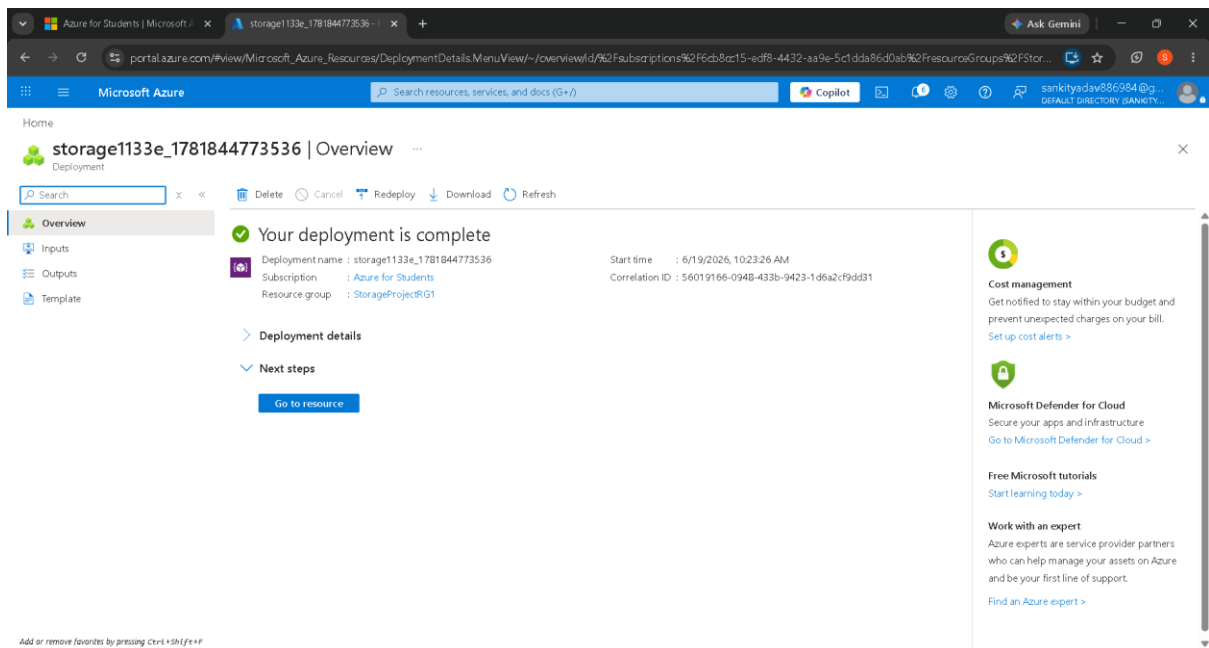


Figure 7:



Step 3 — Create Blob Container

Description

A Blob Container named **projectfiles** was created to manage uploaded files.

Output

Blob Container created successfully.

Figure 8:

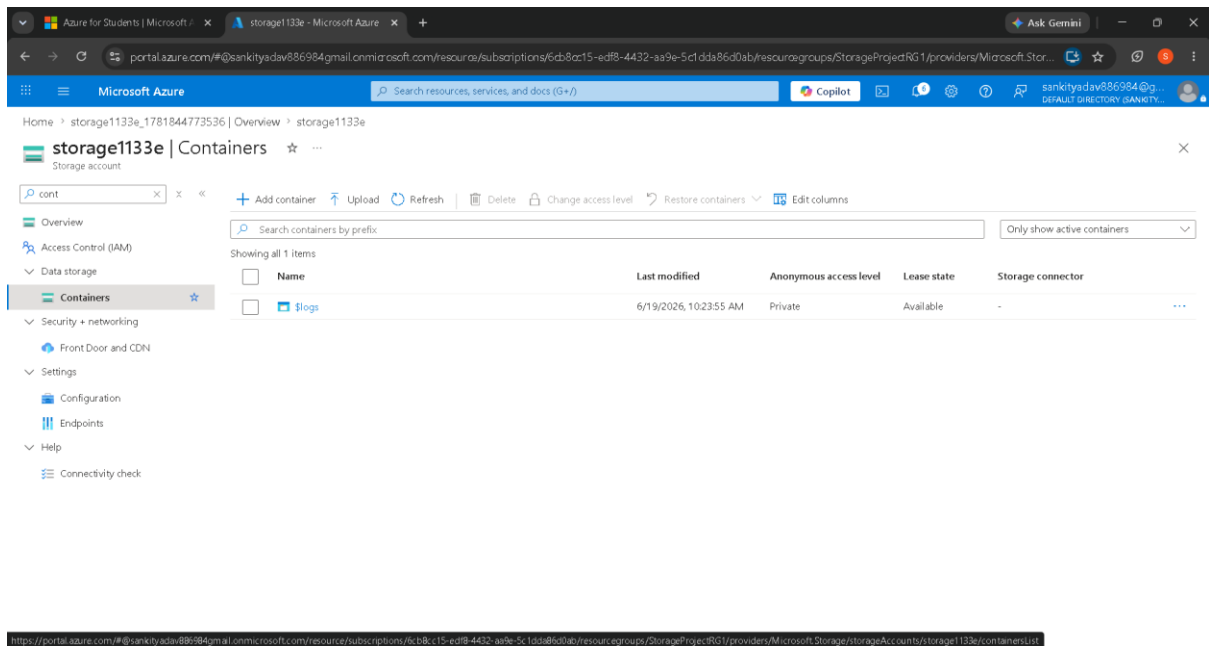
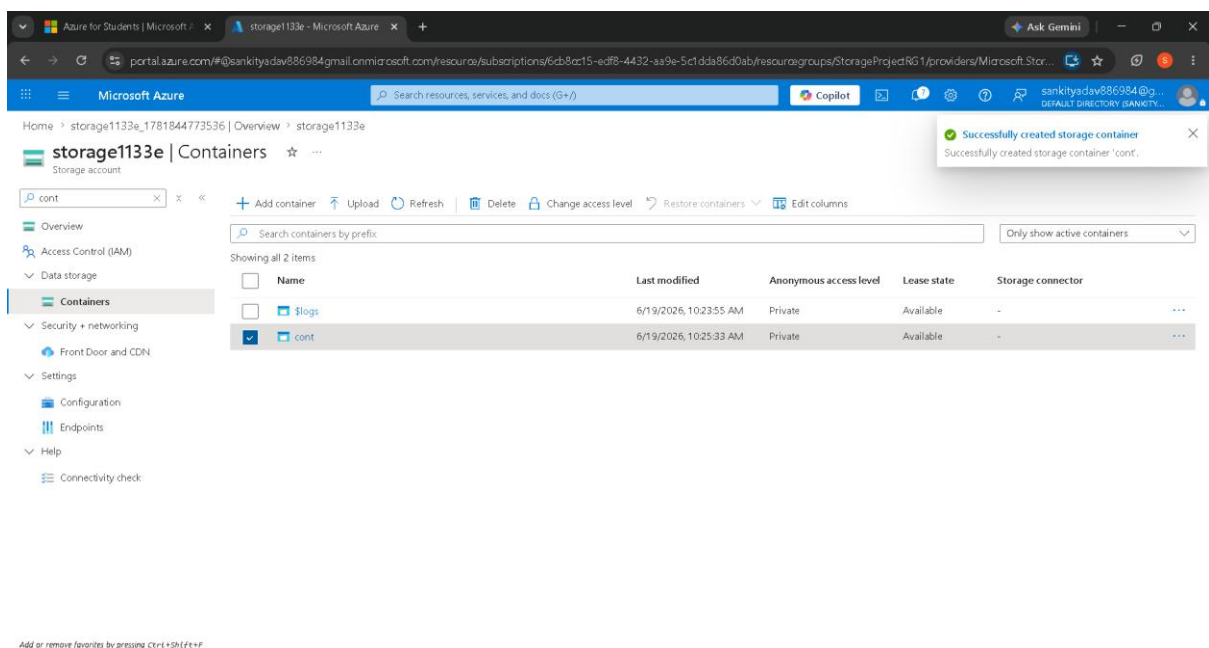


Figure 9:



Step 4 — Upload Files

Description

Files were uploaded to the Blob Container for secure cloud storage.

Output

Files uploaded successfully.

Figure 10:

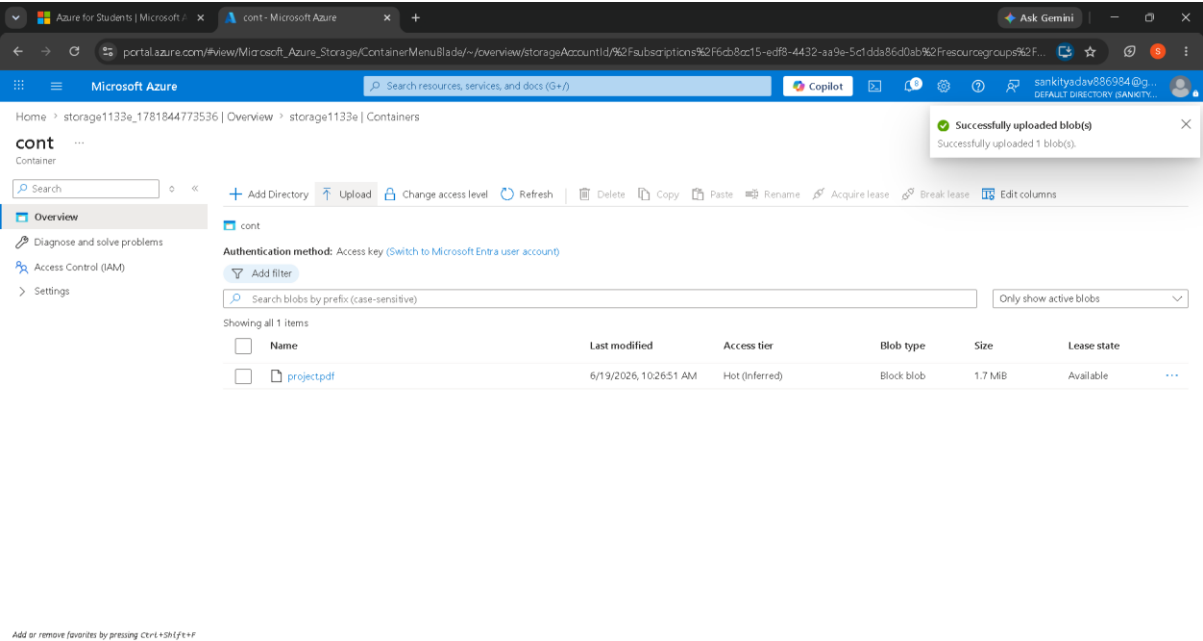
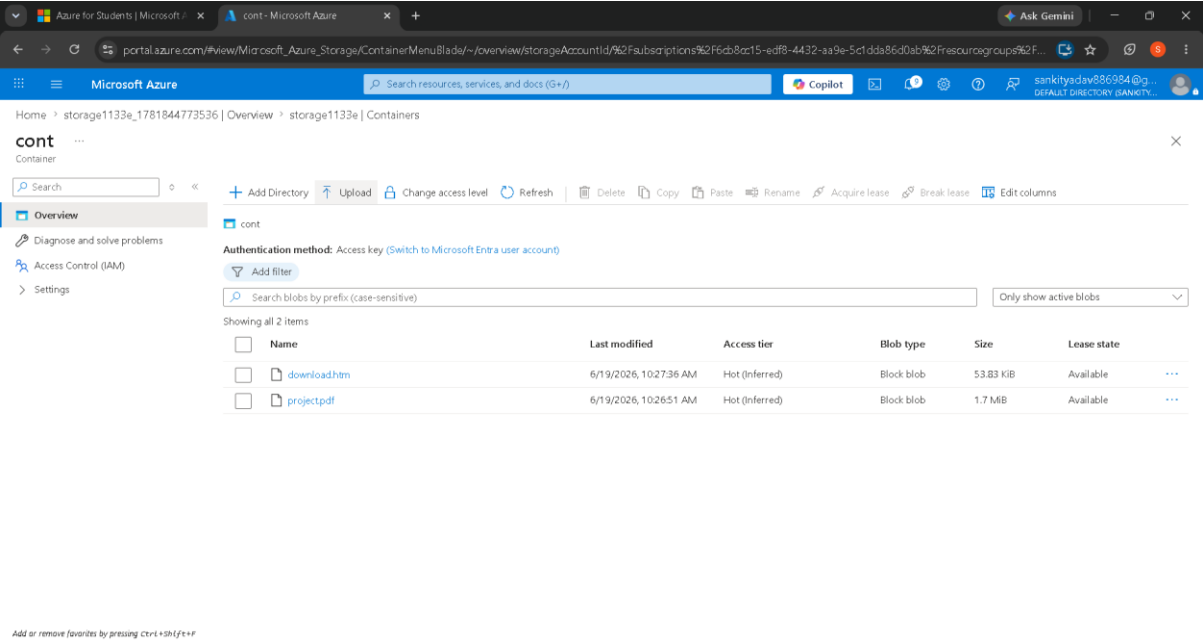


Figure 11:



Conclusion

The project "**Implementation of Cloud File Storage Using Microsoft Azure**" was completed successfully using Azure Storage services. A Resource Group, Storage Account, and Blob Container were created to securely store and manage files in the cloud.

The project provided practical knowledge of cloud storage and resource management using Microsoft Azure. The implemented solution is secure, scalable, and cost-effective for individuals and small organizations.